

ITCS6190 — Assignment #1: Containers with Docker

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1. Exact Commands Used to Build and Run

```
$ make
```

2. Summary Printed to stdout

```
{
  "total_trips": 6,
  "avg_fare_by_city": [
    {"city": "Charlotte", "avg_fare": 16.25},
    {"city": "New York", "avg_fare": 19.0},
    {"city": "San Francisco", "avg_fare": 20.25}
  ],
  "top_by_minutes": [
    {"city": "San Francisco", "minutes": 28, "fare": 29.3},
    {"city": "New York", "minutes": 26, "fare": 27.1},
    {"city": "Charlotte", "minutes": 21, "fare": 20.0},
    {"city": "Charlotte", "minutes": 12, "fare": 12.5},
    {"city": "San Francisco", "minutes": 11, "fare": 11.2},
    {"city": "New York", "minutes": 9, "fare": 10.9}
  ]
}
```

3. Contents of out/summary.json

```
{
  "total_trips": 6,
  "avg_fare_by_city": [
    {"city": "Charlotte", "avg_fare": 16.25},
    {"city": "New York", "avg_fare": 19.0},
    {"city": "San Francisco", "avg_fare": 20.25}
  ],
  "top_by_minutes": [
    {"city": "San Francisco", "minutes": 28, "fare": 29.3},
    {"city": "New York", "minutes": 26, "fare": 27.1},
    {"city": "Charlotte", "minutes": 21, "fare": 20.0},
    {"city": "Charlotte", "minutes": 12, "fare": 12.5},
    {"city": "San Francisco", "minutes": 11, "fare": 11.2},
    {"city": "New York", "minutes": 9, "fare": 10.9}
  ]
}
```

4. Reflection

During this assignment, one difficulty I encountered was fixing a YAML configuration error in `compose.yml`; I used `docker compose config` to validate the configuration before running the stack again. I learned how Docker Compose connects a Python application container to a PostgreSQL container through the service name and how healthchecks and connection retries help avoid database startup problems. I also gained practical experience writing SQL queries in Python and generating a JSON summary from database results. If I improved this project, I would manage database credentials more securely and add more detailed validation and error handling for unexpected database conditions.